

Topological Feature Learning for Parameter Estimation in Neyman-Scott Processes

First-year progression report

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1 Introduction

The study of spatial point processes finds applications in astronomy, cell biology, epidemiology, etc (Møller and Waagepetersen, 2004). Many natural phenomena – such as galaxy clustering – are modelled using SPPs REF, so one can hope to leverage an understanding of these processes to obtain novel insights. The parameters of spatial point processes are of particular interest because they are interpretable scientific quantities. For example, when Neyman-Scott processes are used to model galaxy clustering the parent intensity κ corresponds to the number density of haloes hosting galaxies, the offspring intensity μ is the mean occupancy of a halo, and the cluster scale σ is the spatial extent of the galaxy distribution within a halo; these are exactly the quantities a halo occupancy model is built to constrain REF. In forest ecology, Waagepetersen and Guan (2009) fit Thomas processes to three tropical tree species and read the cluster scale directly as a summary of how tightly seeds are dispersed, finding that species dispersed by exploding capsules, by wind, and by birds and mammals separate along that parameter.

A number of parameter estimation methods have been proposed, typically relying on closed-form expressions for summary statistics, such as the K and L functions (see Section 2.1). Minimum contrast estimation fits parameters by numerically minimizing a discrepancy between the empirical and theoretical curve, most commonly for K or g (Diggle, 2003). Palm likelihood instead builds a composite likelihood from the process’s Palm intensity, and has been developed specifically for Neyman-Scott processes (Tanaka et al., 2008). Fully Bayesian approaches treat the unobserved parent locations as latent variables and sample the posterior over θ by MCMC (Møller and Waagepetersen, 2004).

These methods have three limitations we hope to address with our proposed model. Firstly, they rely on the existence of closed-form expressions of summary statistics which may not exist for every process. This causes two main issues: (i) The expression may not exist: the LGCP-Strauss process of Vihrs (2022) does not have a tractable density, and (ii) The expression may exist but fail the regularity conditions the estimator’s theory needs: Waagepetersen and Guan (2009) require K_ψ to be twice continuously differentiable in ψ , which explicitly excludes the Matérn cluster process.

Secondly, classical parameter estimation methods suffer from hyperparameter sensitivity, specifically the minimum contrast method depends on (r_{max}, q, p) . Naturally, ParamNet also has hyperparameters, but we claim that they are more easily handled and optimized. Minimum contrast’s (r_{max}, q, p) are chosen per dataset, by rule of thumb, with no held-out quantity that tells the practitioner whether the choice was optimal. Waagepetersen and Guan (2009) report that estimates of κ and ω for a Thomas process are “quite sensitive” to the choice of r_{max} and q , and attribute this to a strong negative correlation between $\hat{\kappa}$ and $\hat{\omega}$ – the product $\hat{\kappa}\hat{\omega}$ is stable while the individual estimates are not. ParamNet’s hyperparameters are fixed once on a training split, selected against validation loss, then frozen and applied unchanged to every future cloud.

The choice is therefore made once, with a measurable criterion, rather than repeated per dataset with none.

Finally, classical methods become unstable as the process approaches complete spatial randomness (near-Poisson regime). Let $\lambda = \kappa\mu$ be the expected number of points per unit area, and $\nu = \sigma\sqrt{\kappa}$ the ratio of cluster radius to mean spacing between parents (both are defined properly in Section 3.1). When $\nu \ll 1$ the clusters are well separated and you can practically count them, so κ is easy. As ν grows past 1, neighbouring clusters overlap and the pattern looks Poisson: many (κ, μ) pairs with the same product λ produce nearly the same observable pattern, so κ and μ are only weakly separately identifiable from a single realization. The theoretical K -function makes this explicit – for a Thomas process $K(t) = \pi t^2 + (1 - e^{-t^2/4\sigma^2})/\kappa$, so as σ grows at fixed t the second term vanishes and K tends to the Poisson K regardless of κ (Waagepetersen and Guan, 2009). Minimum contrast is fitted by numerical optimization of a discrepancy that is close to flat along this trade-off, so the estimates it returns in this regime are sensitive to the starting point and to the choice of r_{max} .

Waagepetersen and Guan (2009) note this failure in their simulations: in their weakly clustered case the estimated K could not be distinguished from the Poisson K , and “rather extreme” clustering estimates resulted.

We deliberately retain this regime in our design distribution (see Section 3.1) rather than restricting to well-separated clusters.

These limitations motivate the use of persistence-based features which provide a multi-scale, process-agnostic descriptor of the point cloud that does not require a closed-form or a choice of summary statistic.

1.1 Problem setup

Let X be a Neyman-Scott process on $\mathbb{R}^d$ with unknown parameter vector $\theta \in \mathbb{R}^{n_{params}}$. For example, the homogeneous Thomas process has $\theta = (\kappa, \mu, \sigma)$ corresponding to parent intensity, offspring intensity and cluster scale, respectively. We observe a single realization of X on a bounded window $W \subset \mathbb{R}^d$, giving a point cloud $\mathbf{x} = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$ with $n = |X \cap W|$. The objective is to produce an estimate $\hat{\theta} = \hat{\theta}(\mathbf{x})$ of θ from $\mathbf{x}$ alone. We stress *single realization*: θ is not recovered from repeated draws of X . This is the setting a practitioner faces: one observed pattern and one chance to estimate the generating parameters. It is also what makes the problem hard in a way no estimator can fix: a fixed θ already induces a distribution over point clouds, and it is this irreducible spread – not any weakness of a particular estimator – that sets the error floor characterized below.

The model is trained on a set of synthetic clouds $\{\mathbf{x}_i, \theta_i\}_{i=1}^{N_{train}}$ by backpropagating the RMS loss

$$\mathcal{L}(\hat{\theta}_i, \theta_i) = \sqrt{\sum_{j=1}^{n_{params}} (\hat{\theta}_i^j - \theta_i^j)^2}.$$

The scale of each parameter varies; therefore – if not normalized – the large-scale parameters will dominate this loss. To address this, we train in log-normalized parameter space, mapping $\theta_i \mapsto (\log(\theta_i) - \mu_{\log \theta})/\sigma_{\log \theta}$, where – abusing notation – $\mu_{\log \theta}, \sigma_{\log \theta}$ denote the mean and standard deviation of the logged parameters over the training set.

2 Background

Vihrs (2022) first proposed the use of deep learning for parameter estimation. Yip et al. (2025) employ a similar architecture as well, but use persistence images as the input feature, mapping H_0 – H_2 images of an α DTM ℓ filtration of a halo catalogue to (Ω_m, σ_8) . Their model is designed to estimate cosmological parameters rather than the parameters of the underlying point process.

Furthermore, they do not systematically compare various topological summaries, nor do they address the calibration of the persistence images: they grid-search the kernel bandwidth and raster resolution, but the birth and persistence axis ranges – which fix what a pixel means – are not treated as a design choice at all.

The vectorization of persistence diagrams is an active area of research arising from the issue that persistence diagrams – not being a Hilbert space – cannot be directly used as inputs for machine learning models. To mitigate this, Adams et al. (2017) propose persistence images, which vectorise persistence diagrams by applying a Gaussian kernel to each point in the diagram and integrating over a grid. Umeda (2017) proposes Betti curves, which vectorize persistence diagrams by counting the number of topological features that are alive at each scale. Bubenik (2015) proposes persistence landscapes and Chazal et al. (2014) adapts this idea into persistence silhouettes, both stacking “tent” functions over the diagram points; Reininghaus et al. (2015) avoid vectorization altogether with a kernel on diagrams.

The remainder of this section covers the basic notions in spatial point process and persistent homology that the later sections assume.

2.1 Spatial point processes

Definition 2.1: Intensity function A d -dimensional SPP X on a subset $W \subseteq \mathbb{R}^d$ is equipped with an intensity function $\rho(u) : \mathbb{R}^d \rightarrow \mathbb{R}^+$ such that

$$\mathbb{E}[N(W)] = \int_W \rho(u) du,$$

where $N(W) = |W \cap X|$ denotes the number of points of X in W . When $\rho(u) = \lambda \in \mathbb{R}$, X is said to be *homogeneous*.

Example For a homogeneous Poisson process with intensity λ , we have $\rho(u) = \frac{1}{\lambda}$, i.e. points are scattered uniformly on the space.

Definition 2.2: Neyman-Scott process Let P be a homogeneous Poisson *parent* process with intensity κ . For a Neyman-Scott process with Poisson-distributed offspring we have:

$$\rho_{NS}(u) = \mu \sum_{p \in P} k(u - p),$$

where k is a kernel that determines how the *offspring* points are distributed around the parents.

Note Neyman-Scott processes always have the parent and offspring intensities as parameters. Additional parameters are introduced by the choice of kernel.

Definition 2.3: Homogeneous Thomas process A homogeneous Thomas process is a Neyman-Scott process with an isotropic Gaussian kernel $\mathcal{N}(0, \sigma^2 I)$

$$\rho_{Thomas}(u) = \mu \sum_{p \in P} k(0, \sigma^s),$$

This process has parameters $\theta_{Thomas} = (\kappa, \mu, \sigma)$ corresponding to the parent intensity, offspring intensity, and cluster scale, respectively.

2.2 Persistent homology

Radially averaged statistics – such as the g and K functions – collapse a point cloud to a real-valued curve indexed by scale r . Features based on such statistics are constructed by evaluating over a chosen range $[r_{min}, r_{max}]$ and subsampling at n points along the interval to obtain a feature vector in $\mathbb{R}^n$. This approach is limited because it discards multi-scale connectivity information and void structure, i.e. *how* the density is distributed across the space. Persistent

homology retains this information by tracking a family of simplicial complexes on the point cloud as the scale parameter grows, instead of a single scalar summary. Under a filtration such as Vietoris-Rips, the H_0 persistence summary records the order and scales at which clusters merge. The H_1 counterpart marks voids and gaps in the pattern as they persist across scales, allowing one to capture the clustering-induced emptiness or repulsion that a process may exhibit, which canonical statistics cannot see as clearly due to radial averaging. For each $h \in \mathbb{N}$, the h -dimensional persistence summary requires no closed form expression and no assumption about which process family generated it.

Throughout this section, let X be a point process on $\mathbb{R}^d$ and $W \subseteq \mathbb{R}^d$ a bounded window with $|X \cap W| = n$. Enumerate the points of $X \cap W$ to construct a point cloud $\mathbf{x} = \{x_1, \dots, x_n\} \subset \mathbb{R}^d$. Let $\Sigma(\mathbf{x})$ denote the *full simplex* on $\mathbf{x}$, i.e. the collection of all non-empty subsets of $\mathbf{x}$.

Definition 2.4: Simplex A *simplex* of $\mathbf{x}$ is a non-empty finite subset $\sigma \subseteq \mathbf{x}$. Its faces are the non-empty subsets $\tau \subseteq \sigma$.

Definition 2.5: Simplicial complex A simplicial complex Σ on $\mathbf{x}$ is a collection of simplices closed under taking faces: $\sigma \in \Sigma, \tau \subseteq \sigma \implies \tau \in \Sigma$.

Definition 2.6 A filtration function on $\mathbf{x}$ is a monotone map $f : \Sigma(\mathbf{x}) \rightarrow \mathbb{R}^+$ such that

$$f(\tau) \leq f(\sigma) \quad \forall \quad \tau \subseteq \sigma.$$

Example Vietoris-Rips filtration function:

$$f_{VR}(\{x_i\}) = 0, \quad f_{VR}(\{x_i, x_j\}) = \|x_i - x_j\|,$$

extended to higher simplices by $f_{VR}(\sigma) = \max_{\{x_i, x_j\} \subseteq \sigma} f_{VR}(\{x_i, x_j\})$, i.e. the diameter of σ .

Example For $u \in \mathbb{R}^d$, let $N_k(u)$ denote its k nearest neighbours in $\mathbf{x}$. The DTM_k filtration function is

$$f_{DTM}(\sigma; k) = \max_{u \in \sigma} \left\{ \left(\frac{1}{k} \sum_{v \in N_k(u)} d(u, v)^2 \right)^{\frac{1}{2}} \right\}$$

again extended to higher simplices by the max-over-edges rule of the VR example. (Anai et al., 2020).

Claim 2.7 f_{DTM_k} is monotone, and is hence a valid filtration function (Anai et al., 2020, Prop. 3.5).

Definition 2.8 Given a filtration function f on $\mathbf{x}$ and $r \geq 0$, we construct

$$\mathcal{F}_r = \{\sigma \in \Sigma(\mathbf{x}) : f(\sigma) \leq r\}.$$

Monotonicity of f ensures each $\mathcal{F}_r$ is a simplicial complex and that $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $0 \leq s \leq t$. The family $(\mathcal{F}_r)_{r \geq 0}$ is the *filtration* induced by f . We write $VR_r(\mathbf{x})$ for $\mathcal{F}_r$ under f_{VR} .

Definition 2.9: Betti number Fix $h \in \mathbb{N}$ and a simplicial complex Σ . Taking homology with coefficients in a field (we use $\mathbb{Z}_2$), $H_h(\Sigma)$ is a vector space of dimension $\beta_h = \dim H_h(\Sigma)$, the h -th *Betti number*. $\beta_0, \beta_1, \beta_2, \dots$ count connected components, loops, and cavities, respectively.

Definition 2.10: Persistence pair & lifetime For a filtration $(\mathcal{F}_r)_{r \geq 0}$, a h -dimensional homology class is *born* at the smallest scale $b \geq 0$ at which it appears in $H_h(\Sigma_b)$, and *dies* at the scale $d > b$ at which it merges into an older class or becomes a boundary. The pair (b, d) is a *persistence pair*, and $d - b$ its *lifetime*.

Example Consider H_0 . Every 0-dimensional class is born at $r = 0$ and dies when its two components merge, i.e. at the filtration value of the connecting edge.

Definition 2.11: Persistence diagram A persistence diagram $D_h = \{(b_i, d_i)\}_i$ is the multiset of dimension- h persistence pairs, plotted with birth on the x -axis and death on the y -axis.

Remark Every point of D_h lies above the diagonal $b = d$ because it must die after it is born. Distance from the diagonal is lifetime, so short-lived points near the diagonal are often treated as noise.

Definition 2.12: Tent function For a persistence pair (b_i, d_i) , the tent function is

$$\Lambda_i(t) = \max(0, \min(t - b_i, d_i - t)), \quad t \in \mathbb{R},$$

a triangular bump supported on $[b_i, d_i]$, peaking at height $(d_i - b_i)/2$ when $t = (b_i + d_i)/2$. Persistence landscapes and silhouettes are order statistics and weighted averages of this family; persistence images replace the tent with a smooth Gaussian bump over the same birth-persistence coordinates.

Definition 2.13: Persistence image The persistence image of D_h is constructed by:

1. Mapping to birth-persistence coordinates: $(b, d) \mapsto (b, d - b)$.
2. Weighting each point by $w(d - b)$.
3. Forming the persistence surface $\rho(z) = \sum_i w(d_i - b_i) \phi(z; (b_i, d_i - b_i))$, a weighted sum of isotropic Gaussian kernels with bandwidth σ_{im} centered at the transformed points.
4. Integrating ρ over a 64×64 pixel grid, giving a single-channel image $PI \in \mathbb{R}^{64 \times 64}$.

3 ParamNet

The work completed so far consists of the simulator and its validation, the ParamNet model and its various architectural ablations, comparison with alternative vectorizations, and benchmarking against existing baselines trained on the same data.

3.1 Synthetic Cloud Generation and experimental design

Training data consists of pairs $(\mathbf{x}_i, \theta_i)$ in which $\theta_i = (\theta_i^1, \dots, \theta_i^{n_{\text{params}}})$ is drawn from a design distribution over parameters space and each $\mathbf{x}_i$ is a single realization of the corresponding process. The naive approach of sampling $(\theta_i^1, \dots, \theta_i^{n_{\text{params}}})$ uniformly and independently from per-parameter intervals is inadequate: point processes can typically be separated into qualitative regimes of interest, which occupy curved subsets of the parameter grid. Therefore uniform sampling would allocate most of its mass to configurations that are either near-Poisson or degenerate. We therefore sample in reparameterized coordinates that carry the same meaning for every family considered: the expected offspring intensity $\lambda = \kappa\mu$, the parent intensity κ , and the dimensionless overlap index $\nu = r\sqrt{\kappa}$, where r denotes the cluster scale of the family in question: $r = \sigma$ for Thomas and $r = \sqrt{\sigma_1^2 + \sigma_2^2}$ for nested Thomas. The overlap index ν measures cluster radius against mean inter-parent spacing, so $\nu \ll 1$ gives well-separated clusters and $\nu \gtrsim 1$ gives clusters that merge. Nested Thomas carries an additional coordinate, the scale ratio $\rho = \sigma_2/\sigma_1 \in (0, 1)$. This approach is general to the Neyman-Scott family: every realization has parameters κ and μ by Definition 2.2, and every kernel we consider is a scale family $k(\cdot) = r^{-d}k_0(\cdot/r)$, so it supplies exactly one length. Taking r to be the root-mean-square displacement makes the coordinates comparable across realizations.

We draw $(\log \kappa, \log \lambda, \log \nu)$ uniformly over a grid and invert to recover $\mu = \lambda/\kappa$ and $r = \nu/\sqrt{\kappa}$ for the training labels; log-uniform sampling is used because the parameters span several orders of magnitude. Three constraints are applied to the design grid.

1. *Point budget*: the cost of computing persistence diagrams grows rapidly with the number of points, so we impose $\mathbb{E}[N] = \lambda \in [150, 800]$, fixing the range of $\log \lambda$.

2. *Clusters per window*: we require $\kappa \geq 15$ so that a single realization is informative about κ , and $\kappa \leq 120$ so that individual clusters remain resolvable.
3. *Overlap*: we impose $\nu \in [0.10, 0.90]$, the upper limit deliberately admitting the near-Poisson regime so that the model is still exposed to configurations in which the parameters are weakly identifiable. Parameter draws violating these constraint are rejected and redrawn before any cloud is generated; this filter acts on the design and not on realizations, therefore it does not distort the conditional law of $\mathbf{x}$ given θ .

For a window $W \subset \mathbb{R}^d$, our Neyman-Scott clouds are generated by the standard construction:

1. Expand W by an edge buffer b to produce W_{exp} , so that clusters whose parents lie outside W are not truncated.
2. Sample the parent process on W_{exp} : $n \sim \text{Poisson}(\kappa|W_{exp}|)$, with parents placed uniformly on W_{exp} .
3. For each parent p independently, draw an offspring count $n_p \sim \text{Poisson}(\mu)$.
4. For each offspring independently, draw a displacement $\epsilon \sim k$ and place the point at $p + \epsilon$.
5. Discard the parents and retain the offspring lying in W .

The only element that changes from realization to realization is the displacement kernel k . For the Thomas process $k = \mathcal{N}(0, \Sigma^2 I_d)$; for the Matérn cluster process k is uniform on the ball of radius R ; for nested Thomas the parent process at step 2 is itself a Thomas process rather than a homogeneous Poisson process.

3.2 Method: ParamNet

Let $\mathbf{x}$ denote a point cloud, v its vectorized persistence summary, and $e \in \mathbb{R}^{E+1}$ the resulting feature vector, where E denotes the encoder’s embedding dimension and the $+1$ is the appended $\log N(\mathbf{x})$. ParamNet follows the pipeline

$$\mathbf{x} \xrightarrow{\text{Vectorization}} v \xrightarrow{\text{Encoder}} e \xrightarrow{\text{FCNN}} \hat{\theta}. \quad (1)$$

We note that the pipeline shape follows the neural-estimator patterns used in Vihrs (2022) and Yip et al. (2025). Our contribution lies in the calibration component of the feature-extraction stage, the application to a more challenging process (Nested Thomas), an explicit evaluation of design choices, and comparison of various TDA-based vectorizations. As mentioned in Section 2 there is substantial literature proposing vectorizations of persistence diagrams – persistence images, Betti curves, landscapes, kernel methods – that trade off stability, resolution, and information loss in different ways, but there has been no formalized discussion on which is preferable in this setting.

ParamNet uses the persistence image vectorization, as it is the best performing (Table 3). We now discuss the key components of the model in turn.

Filtration. Instead of the canonical Rips or Čech filtrations, we use the DTM_k filtration (see Example 2.2). Rips is built from inter-point distances alone, making it sensitive to the sparsest connection between two regions and does not register how densely those regions are occupied. DTM_k instead works with a point’s average distance to its k nearest neighbours, providing a more direct measure of local density. Since the parameters of a point process directly affect the local density of the point cloud, we claim that this choice is natural. The scale k can also be fused across several values to give the encoder both fine and coarse views of local density simultaneously; the Thomas model uses a single scale, $k = 5$, while the nested Thomas model fuses $k \in \{5, 10, 15\}$, with the architectural detail deferred to the encoder paragraph below.

Calibration. Fix the homology dimension $h \in \{0, 1\}$; we obtain N_{train} persistence diagrams $\{D_i^{(h)}\}_{i=1}^{N_{train}}$, some of which may have extreme outliers relative to the full set. To ensure that in the resulting persistence images each pixel bears the same meaning, the diagrams must be calibrated to have the same axis ranges. It is not immediately clear that the naive approach of calibrating to the maximum deaths and persistences is adequate in this context: if a diagram in the training set has strong outliers, the remaining diagrams will be rescaled in a way that places the majority of their points relatively close to the origin. Figure 1 demonstrates how this phenomenon might affect the vectorizations.

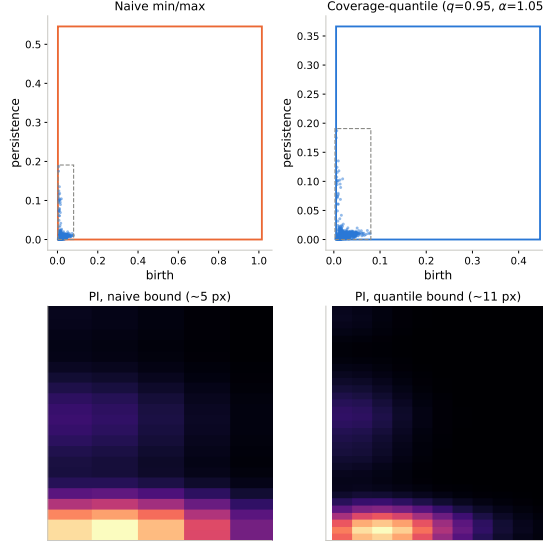


Figure 1: Naive pooled min/max calibration (left) versus per-diagram coverage-quantile calibration, $q = 0.95, \alpha = 1.05$ (right), for the same H_0 diagram (DTM filtration, $k = 5$, Thomas design distribution, Section 3.1). Top: the diagram against its calibrated birth-persistence box; the naive box is stretched by an outlier diagram elsewhere in the population, and the dashed square marks the region zoomed into below. Bottom: that region at each calibration’s native pixel density – a coarse 5×5 block of mostly-fused pixels under the naive bound versus a finer 11×11 grid resolving the secondary peak under the quantile bound, since the same 64×64 grid now covers a narrower range.

Our solution is to calibrate according to *per-diagram coverage quantiles*. For each training diagram $D_i^{(h)}$, define the smallest birth $b_i^{\min}$, largest birth $b_i^{\max}$, and largest persistence value $p_i^{\max}$ *within a single diagram*. Given a coverage level $q \in (0, 1]$ and a padding factor $\alpha \geq 1$, we set the calibrated birth and persistence ranges to be the q -quantiles of these per-diagram statistics, taken over the training set:

$$\left[Q_{1-q}(\{b_i^{\min}\}_{i=1}^{N_{train}}), \alpha \cdot Q_q(\{b_i^{\max}\}_{i=1}^{N_{train}}) \right] \quad \text{and} \quad \left[0, \alpha \cdot Q_q(\{p_i^{\max}\}_{i=1}^{N_{train}}) \right],$$

where $Q_q(\cdot)$ denotes the empirical q -quantile. These two ranges are shared by every diagram of homology dimension h , for every point cloud in the dataset, so that a given pixel of the resulting persistence image always refers to the same region of birth-persistence space. The calibrated ranges are fit *exclusively* on the training diagrams, then frozen and applied unchanged to validation and test diagrams; fitting them on the full, unsplit population of clouds leaks information about the held-out clouds into the training images. For H_0 , birth values are often identically zero under the chosen filtration, in which case the calibrated birth range collapses to a point, and we fall back to a one-dimensional vectorization. The four calibration parameters – coverage q , padding α , kernel width σ_{pixels} , and raster resolution G – are swept one at a time around default values. Informed by the calibration sweep, ParamNet takes $q = 0.95, \alpha = 1.05, \sigma_{\text{pixels}} = 0.5$ and $G = 64$ (values to be re-checked against the run configuration; σ_{pixels} was changed from an earlier 0.75).

Encoder and head. For the Thomas process, we compute the H_0 and H_1 persistence images of the DTM_5 diagram. For Nested Thomas, we compute H_0 and H_1 images for each of the three $DTM_{\{5,10,15\}}$ filtrations and implement a particular architecture to handle the various scales. All images are computed at resolution 64×64 .

For batch size B and $C \in \{1, 3\}$ scales, the encoding stage is

$$(B, C, 64, 64) \xrightarrow{\text{Encoder}} (B, E \cdot C) \xrightarrow{\text{Concatenation}} (B, E \cdot C + 1)$$

In the encoder stage, the image is passed through a small convolutional stack: three 3×3 convolution blocks with channel widths $(32, 64, 128)$, each followed by a ReLU and 2×2 max-pooling with spatial dropout ($p = 0.2$) between blocks. Each block is a CoordConv layer (Liu et al., 2018): two extra channels holding each pixel’s normalized x and y coordinate are concatenated to the image before the convolution, giving the otherwise translation-equivariant stack direct access to absolute position. This matters here because a persistence image’s axes are not arbitrary – birth encodes local density and persistence encodes separation – so a pixel’s location, not just its neighbourhood, is informative. The resulting feature map is reduced by adaptive average pooling and projected by a linear layer to the embedding dimension E (we use $E = 64$). The concatenation step appends $\log N(\mathbf{x})$, the logged number of points in the cloud, to that embedding. The resulting $(B, E + 1)$ tensor is then passed to the FCNN head $NN_\phi : \mathbb{R}^{E+1} \rightarrow \mathbb{R}^{n_{\text{params}}}$, a small feed-forward network with two hidden layers of width 64 and 32, each followed by a ReLU nonlinearity and dropout ($p = 0.1$), then a final linear layer mapping to the n_{params} outputs.

When three scales are used in Nested Thomas, they share one encoder, which works better than using three independent encoders (Table 3, Encoder row). The three resulting feature vectors in $\mathbb{R}^E$ are fused by simple concatenation.

Every experiment uses the same optimizer and schedule: AdamW (lr = 10^{-3} , weight decay 3×10^{-4}), batch size 32, up to 500 epochs with early-stopping patience 50 on the validation loss.

3.3 Evaluation protocol

Data are split by parameter vector so that no parameter value appears in both training and evaluation. Beyond the random test split we generate two further evaluation sets: a grid over the design grid, permitting error to be reported as a surface over the recalibrated parameter space used in Section 3.1; and a set drawn one increment outside each range, characterizing degradation under extrapolation. The classical baselines are run on the same clouds and splits rather than on separate data or numbers taken from the literature.

We report error in the log-normalized parameter space of Section 1.1. For a parameter vector with n_{params} components and a test set of size N_{test} , we define the per-parameter loss

$$\mathcal{L}^j = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (\hat{\theta}_i^j - \tilde{\theta}_i^j)^2 \quad \text{and} \quad \mathcal{L} = \frac{1}{n_{\text{params}}} \sum_j \mathcal{L}^j,$$

where $\tilde{\theta}$ denotes the log-normalized parameter vector and $\hat{\theta}$ the model’s estimate of it.

We measure the *chance-level* by retraining the default PI model with training labels randomly permuted (inputs unchanged). Both processes land at $\mathcal{L} \approx 1.0$ (Table 1), giving an approximate error ceiling against which every other result can be read.

Method	Thomas $\mathcal{L}$	Nested-Thomas $\mathcal{L}$
ParamNet [‡]	0.125 ± 0.007	0.192 ± 0.010
vihrs ($L(r) - r$ CNN)	0.115 ± 0.007	0.217 ± 0.006
Minimum contrast, K	0.855 ± 0.682^a	— ^d
Minimum contrast, g	6.199 ± 0.339^b	— ^d
Chance-level reference (label shuffle)	1.003 ± 0.019	1.006 ± 0.020

Table 1: Estimator comparison on Thomas and Nested Thomas, $n = 10$ seeds, on the shared test split of Section 3.3. [‡]ParamNet is process-specific: DTM₅ for Thomas, fused DTM_{5,10,15} for nested Thomas; see Table 3. ^a Minimum contrast on K is unstable across seeds rather than uniformly bad, and ^b minimum contrast on g is consistently worse than chance; both are discussed in the text. ^dNot run: no closed-form summary statistic for nested Thomas has yet been derived, so its absence is an outstanding experiment rather than an impossibility.

(a) Thomas (κ parent intensity, μ offspring intensity, σ cluster scale)				
Parameter	ParamNet	vihrs	Min. contrast, K	Min. contrast, g
$\mathcal{L}^\kappa$	0.201 ± 0.010	0.192 ± 0.013	1.373 ± 1.087	8.825 ± 0.485
$\mathcal{L}^\mu$	0.134 ± 0.007	0.122 ± 0.007	0.939 ± 0.758	6.334 ± 0.370
$\mathcal{L}^\sigma$	0.040 ± 0.004	0.029 ± 0.002	0.254 ± 0.203	3.437 ± 0.170

(b) Nested Thomas		
Parameter	ParamNet ^c	vihrs
parent intensity	0.283 ± 0.023	0.320 ± 0.010
meta-offspring	0.402 ± 0.022	0.493 ± 0.022
meta-cluster scale	0.143 ± 0.006	0.132 ± 0.009
mean offspring	0.086 ± 0.004	0.110 ± 0.010
cluster scale	0.044 ± 0.000	0.028 ± 0.003

Table 2: Per-parameter loss $\mathcal{L}^j$ underlying the aggregate $\mathcal{L}$ of Table 1 (same methods, same seeds; ParamNet/nested is $n=3$, see that table’s ^c). Minimum contrast has no nested-Thomas columns for the reason given there.

3.4 Estimation performance and baseline comparison

The runs reported below match ParamNet as previously specified on the scale axis: DTM₅ for Thomas, fused DTM_{5,10,15} for nested Thomas. A dagger ([†]) marks an arm on which at least one seed failed to converge and collapsed instead to the $\mathcal{L} \approx 0.9$ – 1.0 chance level; in these cases the reported mean is dominated by the failed seed(s).

We note that training on $\log N$ alone (no topological summary) collapses to the same $\mathcal{L} \approx 0.9$ – 1.0 chance level as the label-shuffle control (on both processes), while removing $\log N$ from the full model costs little on nested-Thomas ($0.192 \rightarrow 0.203$) and is merely noisy on Thomas rather than clearly harmful (both are the “Features” rows of Table 3). Together the two arms show the persistence images – not the point count – are carrying the signal, with $\log N$ a minor, possibly non-essential addition.

We compare ParamNet against the classical and neural baselines of Section 1: minimum contrast fit to Ripley’s K and, separately, to the pair correlation function g , and vihrs – our re-implementation of the neural estimator of Vihrs (2022): a 1-D CNN over the edge-corrected, centred Ripley $L(r) - r$ curve plus $\log N(\mathbf{x})$, i.e. the pipeline of Section 3.2 with the persistence-image feature replaced by a single classical radial summary.

vihrs beats ParamNet on Thomas: its per-parameter losses ($\mathcal{L}^\kappa = 0.192$, $\mathcal{L}^\mu = 0.122$, $\mathcal{L}^\sigma = 0.029$) are each individually lower than ParamNet’s own ($0.201, 0.134, 0.040$; Table 2). So on this single-level process, our topological pipeline does not beat a hand-built radial summary.

On nested Thomas, ParamNet’s gains an advantage on the density/count targets while

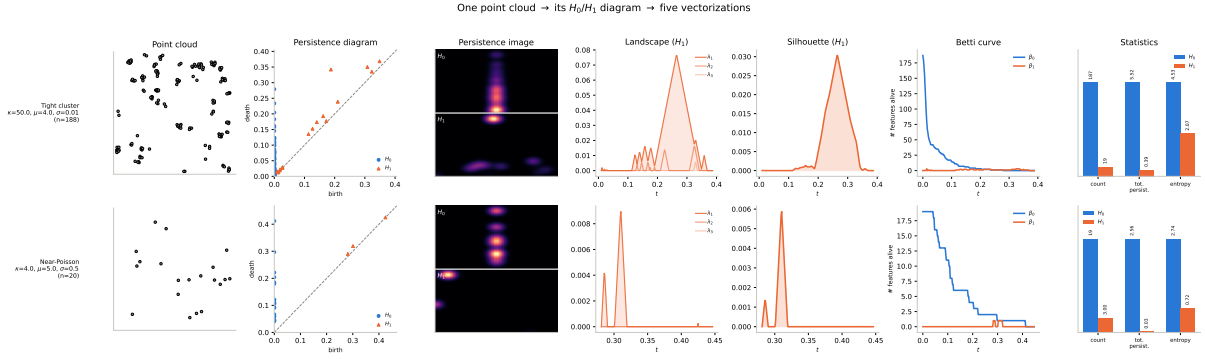


Figure 2: One point cloud $\rightarrow$ its H_0/H_1 persistence diagrams $\rightarrow$ the vectorizations compared in Table 3: persistence-image raster, landscape stack, silhouette curve, Betti curve, and persistence-statistics summary. Top row: the tight-cluster design ($\kappa = 50, \mu = 4, \sigma = 0.01$); bottom row: the near-Poisson design ($\kappa = 4, \mu = 5, \sigma = 0.5$). Statistics-panel bars are min-max normalized per statistic, with raw values printed above each bar, since count, total persistence, and entropy live on incomparable scales.

vihrs maintains an edge on both scale parameters. $L(r) - r$ is built to resolve a characteristic radius directly, while a density-based topological feature is not, and it is only on the more complex two-level process that this trade-off favours our pipeline.

Both neural methods dominate minimum contrast on Thomas by roughly an order of magnitude. The two failure modes are different. On K the estimator achieves losses of 0.365–0.399 for 6/10 seeds – close to *vihrs* – and 0.903–2.056 on the remaining 4/10 – which is the near-Poisson flatness.

On g every seed is worse than chance, indicating systematic bias rather than scattered non-convergence; the likely cause is our contrast exponent $c = 1$, set by analogy with Diggle (2003)’s K -based rule of thumb rather than tuned, where $c = 1/4$ is the value validated for aggregated patterns. We report the number but do not read it as a statement about contrast estimation on g in general.

Parent intensity κ is the hardest target for every method on both processes by a wide margin and the cluster scale σ the easiest, and this ordering holds regardless of which estimator is used; so the difficulty ranking looks like a property of the estimation problem itself rather than an artifact of any one feature set or architecture.

Branches. Table 3 reports the architectural ablations, each measured one axis at a time against the multi-scale, $H_0 + H_1$, CoordConv configuration this work started from (CoordConv itself – see Section 3.2 – is taken as given rather than ablated). Error grows monotonically with k within a single scale on both design distributions: this is consistent with the fact that a finer k -nearest neighbourhood conveys more information about the local-density signal, while a coarser one averages this structure away. None of the three alternative vectorizations beats persistence image on either process or encoder. Figure 2 gives a visual overview of the vectorizations compared, computed from the same persistence diagram.

4 Discussion, limitations and future work

The results support three claims. Persistence images are informative about Neyman-Scott parameters: both processes sit roughly an order of magnitude below the label-shuffle ceiling, and the two “Features” arms of Table 3 place that signal in the images rather than in $\log N$. They are not better than a well-chosen radial summary on a single-level process: *vihrs* wins on all three Thomas parameters, and on both scale parameters of nested Thomas. The advantage is confined to the two-level process and to its density and count targets, which is what one would

Axis	Arm	Thomas $\mathcal{L}$	Nested-Thomas $\mathcal{L}$
Filtration	Rips	— ^a	— ^a
	DTM ₅ / DTM ₁₅	0.125 ± 0.007 / 0.146 ± 0.006	0.205 ± 0.003 / 0.260 ± 0.017
	DTM _{5,10,15} (fused)	0.126 ± 0.008	0.192 ± 0.010^c
Homology	H_0 only / H_1 only	0.125 ± 0.006 / 0.900 ± 0.019	$0.280 \pm 0.250^+$ / 0.932 ± 0.028
Encoder	shared / independent, concat	0.125 ± 0.007 / 0.134 ± 0.009	0.201 ± 0.010 / 0.222 ± 0.021
Vectorization	Persistence image	0.126 ± 0.008	0.192 ± 0.010^c
	Landscape	0.132 ± 0.008	0.250 ± 0.005
	Silhouette ($p = 1$)	0.165 ± 0.013	0.271 ± 0.012
	Betti curve	0.196 ± 0.014	0.290 ± 0.005
Features	log N removed	$0.217 \pm 0.278^+$	0.203 ± 0.007
	Topology removed (log N only)	0.893 ± 0.015	0.948 ± 0.019

Table 3: Branch summary: every axis measured against the multi-scale, H_0+H_1 , CoordConv starting configuration, at the default $q = 0.95$ calibration, $n = 10$ seeds per arm unless noted. Vectorization rows are the native-encoder path. ^a Rips fails on 10/10 seeds before training starts, for the degenerate-birth reason given in the text. ^c $n = 3$, the only seeds run at the current calibration ($\sigma_{\text{pixels}} = 0.5$); see Section 3.2. The DTM₅ / fused rows support the process-specific scale choice of Section 3.2 rather than ablating a single model.

expect of a feature built from local density rather than from a characteristic radius.

Two secondary observations. The ordering of parameter difficulty – κ hardest, cluster scale easiest, for every estimator on both processes – is stable across feature sets and architectures, so it reads as a property of the estimation problem rather than of any one pipeline. And the margin over minimum contrast is not a single effect: on K it is seed-to-seed instability in the near-Poisson regime, on g it is a systematic bias traceable to an untuned contrast exponent. Neither number is a verdict on contrast estimation.

Limitations.

- **Multiple models** The number of filtration scales used is a per-process choice with no established rule beyond experimentation and comparison of validation losses, which undercuts the process-agnostic framing of Section 1. Always fusing would restore a single model at $3\times$ the persistence cost and 0.001 on Thomas, i.e. inside seed noise; attention over scales might be the potential fix.
- **Competitiveness.** Three alternative vectorizations were implemented and none improved on persistence images. As far as we can claim, this is evidence about these summaries under this architecture and this design distribution, not a general claim about vectorizations.
- **No advantage on Thomas.** *vihrs* wins on every parameter of the single-level process, and on both scale parameters of the nested one.
- **Calibration across summaries.** The per-diagram coverage rule was inherited unchanged by the 1-D summaries for comparability, which may not be optimal for each individually.
- **The design distribution is a prior.** The design distribution acts as a prior, so the trained model approximates a posterior summary under it, and its ranges must therefore cover the intended application domain.
- **Computational cost.** Persistence computation scales badly in n ; this is why the point budget of Section 3.1 is capped at 800. Furthermore, the inference cost per cloud is currently unmeasured.

Planned directions.

- The encoder architecture: aim to design a universal architecture that is process-agnostic.

- Inhomogeneous / non-stationary processes, where classical closed forms are least available and the topological approach has the most to gain. The comparator is the two-step procedure of Waagepetersen and Guan (2009), which needs the covariates observed and the intensity correctly specified; a simulation-trained estimator needs neither.
- Apply to real data. Galaxy catalogues if the running example of Section 1 is cosmological. The Barro Colorado Island tree data of Waagepetersen and Guan (2009) might be a second target: a small public data set which is already fitted with a Thomas process.

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